

PY4BIO Mock Exam - practical part

November 20th 2019

Level 1

Write a script that predicts potential cleavage sites by caspases in protein sequences.

Caspases are enzymes that are involved in vital cellular processes and play a prominent role in apoptosis and inflammation. Determining all relevant protein substrates of caspases remains a challenging task. To aid the discovery process you are asked to develop a script to predict caspase cleavage sites in proteins.

Caspase cleavage sites are motifs that always describe an 8-character long peptide. They can be represented by regular expressions that describe the respective compositions of the cleavage sites recognized by the different caspase types.

When a an octapeptide matches the regex, its 4th peptidic bond (ie, between the 4th and 5th letter) will be marked for cleaving or reporting.

For example, the peptide AMQRNLAW is recognized as follows:

A	M	Q	R	*	N	L	A	W
-----	-----	-----	-----	^	-----	-----	-----	-----
P4	P3	P2	P1		P1'	P2'	P3'	P4'

cleavage site

Cleavage rules for the Caspases:

Enzyme name	P4	P3	P2	P1	P1'	P2'
Caspase 1	F, W, Y, or L	-	H, A or T	D	not P, E, D, Q, K or R	-
Caspase 2	D	V	A	D	not P, E, D, Q, K or R	-
Caspase 3	D	M	Q	D	not P, E, D, Q, K or R	-
Caspase 4	L	E	V	D	not P, E, D, Q, K or R	-
Caspase 5	L or W	E	H	D	-	-
Caspase 6	V	E	H or I	D	not P, E, D, Q, K or R	-
Caspase 7	D	E	V	D	not P, E, D, Q, K or R	-
Caspase 8	I or L	E	T	D	not P, E, D, Q, K or R	-
Caspase 9	L	E	H	D	-	-
Caspase 10	I	E	A	D	-	-

Convert these rules into regular expressions covering 8 characters [Hint: the N-glycosylation site detector from Lecture 5] that can be used to search protein sequences for matches.

[Example: the regular expression representing the cleavage rule for Caspase 10 according the instructions is **IEAD.{4}**]

The proteins to be tested are identified by following UniProtKB Swiss-prot accession Ids:

O00238

P17405

Q01814

Q9H3R0

Q9HCP0

Save in a text file (one ID per line).

The easiest way to retrieve the associated sequences is by making usage of the biopython

ExPASy module. Check the documentation on how to use! Remember that what you retrieve is a sequence object, not a raw sequence!

Desired output:

a. Providing a swiss-prot accession ID:

```
> python caspases_cutter.py P17405
```

```
P17405    ASM_HUMAN caspase_1 FLTDLHWD  208  631
```

b. Providing a file with a list of swissprot accession IDs (1 per line)

```
> python caspases_cutter.py test.set
```

```
O00238    BMR1B_HUMAN    caspase_1 LITDYHEN  280  502
P17405    ASM_HUMAN caspase_1 FLTDLHWD  208  631
Q01814    AT2B2_HUMAN    caspase_1 LPADGLFI  225  1243
Q01814    AT2B2_HUMAN    caspase_1 LTTDTSKS  1222 1243
Q9H3R0    KDM4C_HUMAN    caspase_1 YGADINGS  137  1056
Q9H3R0    KDM4C_HUMAN    caspase_1 YGADIIQG  1014 1056
Q9H3R0    KDM4C_HUMAN    caspase_7 DEVGAEV   396  1056
Q9HCP0    KC1G1_HUMAN    caspase_1 LKADTLKE  257  422
Q9HCP0    KC1G1_HUMAN    caspase_1 LFTDLFEK  313  422
```

Meaning of the columns:

column 1: Swiss-prot accession IDs

column 2: Swiss-prot ID

column 3: caspase type

column 4: sequence of recognized cleavage site

column 5: position of cleavage site in the protein sequence

column 6: protein sequence length

Level 2

Build a Python CGI web interface to the caspases cutter tool.

accession	id	caspase	cleavage site	position	protein length
O00238	BMR1B_HUMAN	caspase_1	LITDYHEN	280	502
P17405	ASM_HUMAN	caspase_1	FLTDLHWD	208	631
Q01814	AT2B2_HUMAN	caspase_1	LPADGLFI	225	1243
Q01814	AT2B2_HUMAN	caspase_1	LTTDTSKS	1222	1243
Q9H3R0	KDM4C_HUMAN	caspase_1	YGADINGS	137	1056
Q9H3R0	KDM4C_HUMAN	caspase_1	YGADIIQG	1014	1056
Q9H3R0	KDM4C_HUMAN	caspase_7	DEVDGAEV	396	1056
Q9HCP0	KC1G1_HUMAN	caspase_1	LKADTLKE	257	422
Q9HCP0	KC1G1_HUMAN	caspase_1	LFTDLFEK	313	422

For the evaluation of the practical part of the exam we apply following scoring principles:

Execution (6 pt)

The source code is executed with given input and we check whether the output generated by the script corresponds with the expected output (5 pt)

Did the student correctly translated the problem statement into a series of sequential steps describing what the program must do (1 pt).

Programming style (4 pt)

Is the source code well structured? (1 pt)

Is the source code well documented? (relevant/neither too much or too little) (1 pt)

Is the naming of the different components of a program consistent? (1 pt)

Did the student use functions and/or methods? If so, are they not too long? (divide-and-conquer principle) (1 pt)